ROOT CAUSE UNLOCKED · CONDITION KIT

The Heart and Metabolic Root Cause *Kit*

Which numbers actually predict risk, which ones mislead, and exactly what to ask your doctor for

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The Heart and Metabolic Root Cause Kit

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BEFORE YOU START

Contents

Before anything else: when to stop reading and call 911

Call 911 now if you have:

- Chest pain, pressure, tightness, or squeezing, especially with sweating, nausea, breathlessness, or pain spreading to the jaw, neck, back, or arm. **Do not drive yourself.**
- Sudden weakness or numbness on one side, face drooping, trouble speaking or understanding, sudden vision loss, or sudden severe unexplained headache. Note the time symptoms started, because treatment decisions depend on it.
- Sudden severe breathlessness, or coughing pink frothy sputum.
- A tearing or ripping pain in the chest or back.

Women, and people with diabetes, more often present without classic crushing chest pain. Unexplained breathlessness, unusual fatigue, nausea, or jaw and back discomfort can be the presentation. If something feels badly wrong and new, that is enough reason to be seen.

This section is first on purpose. Everything else in this kit is for a calm day.

Read this before you read anything about statins

Key 3 of *Unlock Your Heart Health* is titled around what the statin evidence actually shows. This kit takes the same position, and I want it clear from the first page:

Nothing in these guides is a reason to stop a prescription on your own. Stopping a cardiovascular medication on the strength of a wellness document is one of the few things in this entire product line that could genuinely harm you.

What the supplement guide does instead is give you the whole picture, including the parts both sides leave out: that industry-funded drug research is measurably more favourable to the sponsor, that **in true primary prevention lipid-lowering therapy has not been shown to reduce mortality**, and that the muscle-symptom objection does not survive blinded testing. Those three things are all true at once, and knowing all three is what lets you have a real conversation instead of a one-sided one.

The nine keys, and what tests each one

Key	What it means	How you find it
1. Insulin resistance	The shared root of cholesterol, blood pressure, and fatty liver	Fasting insulin, glucose, HbA1c, triglyceride to HDL ratio
2. Beyond the lipid panel	What a standard panel leaves out	**ApoB and Lp(a).** The highest-yield additions here
3. Statin evidence	What it shows and does not	Covered in the supplement guide
4. Inflammation	The blood vessel lining	hs-CRP
5. Liver	A metabolic organ, not just an alcohol story	ALT, AST, GGT, imaging, fibrosis scoring
6. Blood pressure	Bigger than salt	A validated home monitor. Not a blood test
7. Visceral fat	Where you store it matters	A tape measure. Not a blood test
8. Sleep	The underrated driver	Sleep apnea assessment, sleep log
9. Nutrition and movement	What has evidence behind it	The diet guide

Three of the nine are not blood tests at all, and one of those, a tape measure around your waist, may tell you more than most of the panel.

Key 2: the two tests your standard panel probably left out

ApoB

What it is. Every atherogenic particle carries exactly one apolipoprotein B molecule. So ApoB counts the particles, while LDL cholesterol measures the cholesterol carried inside them. Those are different, and they come apart in a specific group of people.

Why it matters. A review comparing ApoB with non-HDL cholesterol as a treatment target concluded that ApoB is preferred as a secondary target in people with mild-to-moderate high triglycerides (175 to 880 mg/dL), diabetes, obesity or metabolic syndrome, or very low LDL cholesterol under 70 mg/dL, and that where ApoB is unavailable, non-HDL cholesterol should be used to supplement LDL cholesterol (PMID 32562186).

Read the list of who that applies to. Diabetes, obesity, metabolic syndrome, high triglycerides. That is precisely the population Key 1 of this book is about. If insulin resistance is your picture, your LDL cholesterol can look reassuring while your particle count does not.

What to ask for. ApoB. If your lab does not offer it, ask for **non-HDL cholesterol**, which is simply total cholesterol minus HDL and can be calculated from the panel you already have.

Lipoprotein(a)

What it is. A genetically determined lipoprotein, largely set by inheritance and barely moved by diet or lifestyle.

Why it matters. Lp(a) is an independent risk factor for atherosclerotic cardiovascular disease including acute coronary syndrome, peripheral arterial disease and stroke, and for calcific aortic stenosis, with an estimated 20% of the population affected (PMID 41544761).

The practical points.

- **Measure it once in your life.** It is largely genetic and does not need repeating.
- **A high result does not mean you failed at lifestyle.** It means your baseline risk is higher and everything else you can control matters more.
- **It has family implications.** If yours is high, first-degree relatives are worth testing.
- Targeted therapies are in development, and current management focuses on controlling everything else more aggressively.

Key 1: insulin resistance, the shared root

The ebook's argument is that cholesterol, blood pressure and fatty liver share one engine. Test the engine.

What to ask for: fasting insulin, fasting glucose, and HbA1c.

And one ratio you can calculate yourself right now, from a panel you probably already have: **triglycerides divided by HDL**. It costs nothing, it is not a validated diagnostic threshold, and it is a useful pointer toward the insulin-resistant pattern when the ratio is high. Bring it to your appointment as a question rather than a conclusion.

Why fasting insulin matters. Glucose stays normal for years while insulin climbs to hold it there. Measuring only glucose means finding the problem after the compensation fails, which is exactly the window you want to act inside.

Key 4: inflammation

hs-CRP, the high-sensitivity version, not standard CRP. Standard CRP is built to detect infection-level inflammation; the high-sensitivity assay reads the low range that matters for vascular risk.

One recent paper describes bundled measurement of LDL cholesterol, Lp(a), and high-sensitivity CRP across the risk spectrum, which is a reasonable way to think about ordering all three together rather than piecemeal (PMID 41509673).

How to read it. A single raised hs-CRP after a cold means nothing. Repeat it when you are well before drawing conclusions.

Homocysteine: the marker that needs the most honest handling in this kit

What it is. An amino acid your body produces and clears in the methylation cycle, a set of reactions that runs on B12, folate, and B6. It backs up in the blood when one of those vitamins runs short, when kidney or thyroid function is off, or when a common MTHFR gene variant slows the recycling enzyme.

What an elevated result actually means. Treat it as a signal to investigate B12, folate, B6, thyroid, and kidney status, not as a cardiac target proven to change outcomes on its own. That distinction is the whole game with this marker, and a lot of wellness content gets it backwards.

The honest evidence, in three sentences. Observationally, lower homocysteine tracks with modestly lower heart disease and stroke risk, "at most a modest independent predictor" in the pooled data (PMID 12387654). Genetically, lifelong moderate elevation showed little or no effect on coronary disease (odds ratio 1.02) once 19 unpublished datasets were included, with the published literature's positive result attributed to publication bias (PMID 22363213). And in randomized trials, lowering it with folic acid cut stroke by about 10 percent, mostly in folate-deficient populations, while coronary disease did not move (PMID 27528407).

Notice which direction that bias ran. The homocysteine literature skewed toward the position wellness culture favors, exactly as industry funding skews drug literature toward the sponsor's position. This kit applies the same test to both, and reports what survives it.

Which range frame you are read against matters. At Root Health I read against a functional target under 7 umol/L. Standard laboratory ranges run to 10 or 15 depending on the lab. A result above roughly 10 is worth the workup conversation either way.

If it comes back elevated, the follow-up panel is: B12, folate, B6, TSH, and kidney function (creatinine and eGFR). MTHFR genotyping is optional; it changes which vitamin form makes sense, not the conclusion.

One boundary. Everything above concerns moderate elevation. Homocystinuria, the rare inherited disorder where levels run several times normal, is a genuinely dangerous medical condition, and none of the reassurance above applies to it.

Key 5: your liver

Fatty liver is now usually called metabolic dysfunction-associated steatotic liver disease, or MASLD, and the name change is the point: it is a metabolic condition, not primarily an alcohol story.

What to ask for. ALT, AST, and GGT to start. A normal ALT does not exclude fatty liver, which is the most common misunderstanding here. If there is suspicion, imaging and a non-invasive fibrosis score are the next step, and that is a physician's call.

Why it is worth finding. A systematic review and network meta-analysis of randomized trials found sustained weight loss associated with histological improvement in MASLD in people with obesity, while being explicit that the limited evidence base and sparse network preclude causal inference and that the findings should be treated as exploratory and hypothesis-generating (PMID 41804193).

That is a fair summary of where the field is: weight loss is the lever, and the trial base is thinner than the confidence with which it is usually stated.

Keys 6 and 7: the two measurements that are not blood tests

Blood pressure

Buy a validated upper-arm home monitor. Wrist cuffs are less reliable. A single reading in a clinic is a poor measure of anything, and home readings across a week are far better.

How to take it properly, because most home readings are taken wrong: sitting, back supported, feet flat, arm supported at heart height, no talking, after five minutes of quiet, and not within thirty minutes of caffeine, exercise or smoking. Two readings a minute apart, morning and evening, for seven days.

Your waist

A tape measure around your waist at the level of the navel tells you about visceral fat, which is Key 7 and is more metabolically active than fat elsewhere. Measured over a year, it is a better progress signal than the number on the scale, because it moves when body composition changes even if weight does not.

Functional ranges for the markers in this guide

Clinical targets I use at Root Health, not validated diagnostic thresholds, narrower than the standard laboratory range on purpose. **Lipid targets are deliberately absent from this table**, because the right LDL or ApoB target depends on your overall risk, your history, and whether you are in primary or secondary prevention, and a one-size number would mislead. That is a conversation with your physician using a risk calculator.

Marker	Units	Functional target	Approximate standard range
Glucose, fasting	mg/dL	75 to 86	70 to 99
HbA1c	%	4.6 to 5.3	4.0 to 5.6
hs-CRP	mg/L	under 0.5	under 1.0
Ferritin	ng/mL	50 to 150	25 to 300
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
Homocysteine	umol/L	under 7.0	under 10 to 15, varies by lab
Magnesium (RBC)	mg/dL	5.2 to 6.8	4.0 to 7.2
Albumin	g/dL	4.2 to 4.8	3.5 to 5.0
White blood cells	K/uL	5.0 to 7.5	3.5 to 10.5

Reading your results as a pattern

Pattern one: the metabolic cluster. High triglycerides, low HDL, creeping fasting insulin, raised ALT, expanding waist, blood pressure drifting up. That is one problem wearing five labels, and Key 1 is the answer.

Pattern two: reassuring LDL, unreassuring ApoB. Common in exactly the group named in the ApoB review above. The particle count is the number to act on.

Pattern three: everything good except Lp(a). Genetic, not your fault, and it raises the value of controlling everything you can control.

Pattern four: normal panel, real symptoms. If you have exertional chest discomfort or breathlessness, a normal lipid panel does not clear you. That is a cardiology question, not a supplement question.

The conversation with your doctor

A request list you can bring

- Standard lipid panel, plus **ApoB**, or non-HDL cholesterol if ApoB is unavailable
- **Lp(a), once in your life**
- Fasting insulin, fasting glucose, HbA1c
- hs-CRP
- Homocysteine, read as a B-vitamin, thyroid, and kidney workup trigger, not a cardiac target
- ALT, AST, GGT
- A cardiovascular risk estimate using a validated calculator, with your numbers in it
- Sleep apnea assessment if you snore or wake unrefreshed

The sentence that helps

“My LDL looks fine but I have metabolic syndrome. Could we check ApoB, since I understand LDL cholesterol can underestimate particle count in that setting?” Specific, accurate, and it opens the right conversation.

What these labs cannot tell you

- **Whether you have plaque.** Lipids estimate risk. Imaging such as a coronary calcium score answers a different and more direct question, and is worth discussing with your physician.
- **Your risk on their own.** Risk is calculated from a combination, not read off a single line.
- **Whether you need a statin.** That is a risk-based decision with your physician, covered in the supplement guide.

When to seek clinical review

Emergency, repeated from the top: chest pain or pressure, one-sided weakness or speech trouble, sudden severe breathlessness, tearing chest or back pain. Call 911.

Soon: blood pressure repeatedly at or above 140/90 at home, chest discomfort on exertion that eases with rest, new breathlessness on stairs you used to manage, swelling in both ankles, or an ALT that stays raised.

Your next step

The companion guides do the rest: **The Heart and Metabolic Diet Guide** covers the eating pattern with the best blood pressure and metabolic evidence, and **The Heart and Metabolic Supplement Guide** covers what genuinely helps alongside your medications, including the honest read on red yeast rice.

> ***Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=heart-labs***

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

> ***Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=heart-labs***

If you would rather start on your own, start with home blood pressure readings and a waist measurement. Those two cost almost nothing and they move:

> ***Free tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=heart-labs***

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All references were verified against live PubMed records on 2026-08-09, including a publication type and corrections check for retractions and errata. Verification detail is documented in `Heart-Kit-Citation-Ledger.md`.

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Educational content only. Not medical advice. Not a diagnosis. Not a reason to stop or refuse any cardiovascular medication. Chest pain is a medical emergency until proven otherwise.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Call 911 for chest pain or pressure, especially with sweating, nausea, breathlessness, or pain into the jaw, neck, back or arm; for sudden one-sided weakness, face droop, or trouble speaking; for sudden severe breathlessness; or for tearing chest or back pain. Do not drive yourself.

Nothing here is a reason to stop or refuse a statin, a blood pressure medication, or metformin.

If you have kidney disease, heart failure, or take an ACE inhibitor, ARB, or potassium-sparing diuretic, **talk with your clinician before increasing potassium or using a potassium-based salt substitute.** That caution appears again where it matters below, and it is the one item in this guide that can hurt the wrong person.

The one idea underneath all three conditions

Key 1 of *Unlock Your Heart Health* argues that high cholesterol, high blood pressure, and fatty liver share a root: insulin resistance. That is why this is one diet guide rather than three.

It also means the food work is not primarily about avoiding dietary cholesterol or counting fat grams. It is about the pattern that improves insulin sensitivity, blood pressure, and liver fat at the same time. Those move together, and so do the interventions.

Part one: what actually lowers blood pressure

This is the section with the best evidence-to-effort ratio in the whole kit.

What a network meta-analysis found. In adults ranging from prehypertension to established hypertension, comprehensive lifestyle modification, aerobic exercise, isometric training, **low-sodium and high-potassium salt**, salt restriction, breathing-control, meditation, and a low-calorie diet all showed clear effects on blood pressure reduction (PMID 32975166).

Note what is on that list. Not just salt restriction. Breathing and meditation are on it too, which is unusual for a cardiology paper and worth taking seriously.

The sodium and potassium point, which is better than “eat less salt”

A systematic review and meta-analysis found that a lower 24-hour urinary sodium to potassium ratio is associated with lower blood pressure in adults, and concluded that dietary strategies raising potassium while lowering sodium would be beneficial for blood pressure (PMID 34117485).

That reframes the task usefully. It is a ratio. You can improve it from either end, and adding potassium-rich food is easier to sustain than joylessly cutting salt.

Potassium-rich foods: potatoes with skin, beans and lentils, leafy greens, tomato products, avocado, banana, citrus, yogurt, salmon.

The safety line, repeated because it matters: if you have kidney disease or heart failure, or take an ACE inhibitor, ARB, or potassium-sparing diuretic, **do not increase potassium or use a potassium salt substitute without your clinician**. In those situations high potassium is genuinely dangerous.

Where the sodium actually is. Not the shaker. Bread, deli meat, cheese, sauces, soups, and restaurant food carry most of it. Cooking at home more often does more than removing the shaker ever will.

The DASH pattern, with the honest evidence grade

DASH is the most recommended eating pattern for blood pressure. A 2026 systematic review and meta-analysis comparing DASH against alternative structured dietary interventions found benefits on blood pressure and selected cardiometabolic parameters, while stating plainly that **the overall certainty of evidence was low with substantial heterogeneity**, and that findings should be interpreted cautiously (PMID 42492868).

How to read that. DASH is a sensible, well-tested pattern. It is not clearly superior to other structured healthy patterns, and the comparative evidence is weaker than its reputation. So if you would rather eat a Mediterranean pattern, that is a reasonable choice, and adherence matters more than which structured pattern you pick.

The DASH shape, plainly: vegetables and fruit at most meals, whole grains, beans, nuts and seeds, low-fat dairy if you eat dairy, fish and poultry over red and processed meat, and less added sugar and sodium.

Part two: eating for insulin, which is the shared engine

The same five practices I use in every insulin-driven picture, because the engine is the same one:

1. **Protein at every meal, starting with breakfast.**
2. **Never eat carbohydrate naked.** Pair it with protein, fat, or fiber.
3. **Fiber at every meal**, and see the soluble fiber note below.
4. **Cut liquid sugar first.** Sweetened drinks and juice.
5. **Walk after meals**, ten to fifteen minutes.

Sourcing note. Those five are standard clinical nutrition for insulin resistance rather than findings from a cardiovascular outcome trial, and no citations are attached to them for that reason. The blood pressure and liver claims in this guide are cited.

Soluble fiber deserves a specific mention because it does two jobs at once here: it blunts the post-meal glucose rise and it lowers LDL cholesterol modestly and reliably. Oats, barley, beans, lentils, psyllium, apples, and citrus. Build up gradually with plenty of water.

Part three: the liver, which is Key 5

Fatty liver, now usually called metabolic dysfunction-associated steatotic liver disease, is a metabolic condition rather than primarily an alcohol story.

Weight is the lever. A systematic review and network meta-analysis of randomized trials found sustained weight loss associated with histological improvement in MASLD in people with obesity, while being explicit that the limited evidence base and sparse network preclude causal inference, and that the findings are exploratory and hypothesis-generating (PMID 41804193).

Coffee, which is the pleasant surprise in this guide. An umbrella review plus systematic review and meta-analysis of coffee and non-alcoholic fatty liver disease found mixed results for preventing NAFLD in the general population, but **benefits of coffee consumption on liver fibrosis among patients who already have NAFLD** (PMID 34966282).

Read that split precisely: coffee is not established as preventing fatty liver, and among people who have it there is a signal on fibrosis. Black or minimally sweetened, since a coffee drink carrying 40 g of sugar is working against the rest of this guide.

Fructose deserves naming. Liquid fructose in particular, meaning sweetened drinks and juice, is handled largely by the liver. This is the same recommendation as cutting liquid sugar above, arriving from a different direction, which is usually a sign it is worth doing.

Alcohol. Even in the metabolic form of fatty liver, alcohol adds a second injury to a liver already under load. If you have fatty liver, reducing it is a direct intervention rather than general advice.

Part four: a week that works

Breakfast. Oats with nuts and berries. Eggs with vegetables and beans. Plain yogurt with fruit and seeds.

Lunch. Lentil or bean soup with wholegrain bread. Large salad with fish or chicken, olive oil, and a fiber-rich carbohydrate.

Dinner. Fish two or three times a week. Otherwise poultry, beans, or lentils, with two vegetables and a starch such as potato with the skin on, which is also a potassium win.

Snacks. Nuts, fruit, yogurt. Pair rather than eat carbohydrate alone.

Drinks. Water, coffee, tea. This is the single largest change most people can make and it hits blood pressure, insulin, and liver at once.

Cooking at home more often is the mechanism behind most of the sodium reduction above, and it does not require a recipe list.

What I deliberately left out, and why

- **Blanket dietary cholesterol restriction.** For most people, dietary cholesterol is a much smaller lever than the pattern as a whole. Eggs are not the problem in most diets.
- **Very low fat diets.** Long out of step with the evidence, and hard to sustain.
- **Ketogenic diets as the default.** They can improve triglycerides and liver fat, and their effect on LDL is variable and sometimes markedly adverse. If you want to try one with a raised LDL or ApoB, that needs monitoring and a physician, not a guide.
- **Grapefruit, if you take a statin or several blood pressure medications.** A genuine interaction, and worth checking against your specific prescription.
- **Potassium salt substitutes for everyone.** Genuinely useful for many people and genuinely dangerous in kidney disease, heart failure, and on certain medications. This is a clinician question.
- **Detox and cleanse protocols for the liver.** No outcome evidence, and the actual intervention with evidence is weight, alcohol, and sugar.

Telling you what to skip is part of what you paid for.

Running a fair experiment

Weeks 1 and 2, baseline. Home blood pressure twice daily, waist measurement, and the labs from the lab guide.

Weeks 3 to 10, the pattern. Sodium down and potassium up, protein at breakfast, soluble fiber daily, liquid sugar out, walk after meals.

Week 11, measure. Blood pressure responds within weeks. Lipids and liver enzymes need eight to twelve weeks, so time your retest accordingly.

Realistic expectations

What is realistic. Meaningful blood pressure change within four to eight weeks from the sodium, potassium, movement and weight work. Modest LDL reduction from soluble fiber. Real liver improvement over months with sustained weight loss.

What is not realistic. Diet alone normalizing a genetically high Lp(a), or replacing appropriate medication in someone at high calculated risk. Those are additive, not alternatives.

When to seek professional care

Emergency: chest pain or pressure, one-sided weakness or speech difficulty, sudden severe breathlessness, tearing chest or back pain. Call 911.

Soon: home blood pressure repeatedly at or above 140/90, chest discomfort on exertion, new breathlessness on stairs, swelling in both ankles, or a persistently raised ALT.

Working with Root Health

This guide is written to be safe and useful for a reader I have never met. A personalized plan starts from your numbers, your calculated risk, and your medication list.

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Read the companion documents alongside this one. **Understanding Your Heart and Metabolic Labs** covers ApoB and Lp(a), and **The Heart and Metabolic Supplement Guide** covers red yeast rice honestly, including what it actually contains.

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Nothing in this guide is a reason to stop a prescription on your own. The statin section below gives you the honest evidence in both directions, including who funded it, so you can have a real conversation with your prescriber instead of a one-sided one.

Supplements in the United States are regulated as food, not as drugs. Quality varies. Look for independent third-party testing.

How to read “no evidence” in this guide

The statin section below explains how funding shapes research results in this field. The same economics shape what never gets studied at all: a large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back, which nobody can do with magnesium or an herb. So when this guide says “no evidence,” it will tell you which of three things it means: *tested and failed* (a real trial ran and the answer was no), *never properly tested* (nobody could afford the trial, so absence of evidence is not evidence of absence), or *tested small and won* (real but modest trials, usually on markers rather than outcomes, because markers are what a small budget can afford).

The asymmetry earns nutrients humility, not automatic credit: small nutrient trials carry their own publication bias toward positive findings, just as drug trials tilt toward their sponsor (PMID 28207928). One rule, applied in both directions: name the funding, name the evidence tier, and let you see both.

Statins: the honest version, including the parts the drug companies do not advertise

Most of the functional medicine world is statin-skeptical, and a lot of that skepticism is well founded. Some of it is not. Sorting the two is worth more to you than a slogan in either direction, so here is the whole picture.

First, the criticism that is completely valid: who paid for the research

Industry-funded drug research produces more favourable results than independently funded research, and this is a documented, quantified effect rather than a suspicion. A Cochrane methodology review found that sponsorship of drug and device studies by the manufacturing company leads to more favourable efficacy results and conclusions than sponsorship from other sources, and concluded that this reflects **an industry bias that cannot be explained by standard risk-of-bias assessments** (PMID 28207928).

That last clause matters enormously. It means the usual quality checks a reviewer applies do not catch it. So “this was a well-conducted randomized trial” is not a sufficient answer to “who funded it.”

The large statin efficacy trials, the ones establishing cardiovascular benefit, were overwhelmingly manufacturer-funded. Anyone who tells you that fact is irrelevant is not being straight with you.

Second, the distinction that changes everything: primary versus secondary prevention

This is the single most important thing in this guide and it is routinely collapsed in both directions.

Secondary prevention means you have already had a heart attack, a stroke, or documented cardiovascular disease. The evidence for statins here is strong and consistent, and this is not seriously disputed.

True primary prevention means you have none of that and are being treated on the basis of estimated risk. A 2026 meta-analysis of randomized trials looked specifically at people **without** prior cardiovascular, vascular, or cerebrovascular disease, because earlier analyses had blurred the line by including people who already had disease. In that genuinely primary-prevention population, lipid-lowering therapy **did not reduce all-cause mortality or cardiovascular mortality**, while it did significantly reduce **nonfatal** cardiovascular events and cardiovascular hospitalizations. The authors state that additional research is needed to define the benefits in true primary prevention (PMID 41864911).

Read that carefully, because it is the crux of the whole argument. In true primary prevention, the honest claim is a reduction in nonfatal events, not a demonstrated reduction in dying. That is a real benefit and it is a materially smaller claim than most people are given. If you are a healthy 52-year-old with a moderately raised LDL and no disease, that finding is about you, and you are entitled to know it before you decide.

Third, the criticism that does not hold up: muscle symptoms

Here the skepticism runs ahead of the evidence, and I checked the funding on these specifically because it is a fair question to ask of anything I cite.

StatinWISE was a series of **200 N-of-1 randomized controlled trials** in primary care, in which individual patients alternated blinded between statin and placebo. It was **funded by the UK National Institute for Health Research and the Department of Health, and run from the London School of Hygiene and Tropical Medicine. Not industry.** Most participants who completed the trial decided to continue taking statins afterwards (PMID 33709907).

A separate systematic review and meta-analysis of blinded statin rechallenge, from a university family medicine department declaring no competing interests, found **no significant difference in mean myalgia or global symptom score between statin and placebo**, that only about **one-third** of people previously believed to be statin intolerant were genuinely unable to tolerate a statin on blinded rechallenge, and that **one-quarter were intolerant of placebo** (PMID 38128013).

This does not mean anyone's symptoms are imaginary. Symptoms are real and felt. It means that attributing them to the statin without blinding is unreliable, and that most people who believe they cannot take one can.

So what should you actually do

The useful question is not "are statins good or bad." It is **"what is my actual risk, and what does the evidence say at that level of risk."**

1. **Get your risk calculated** with a validated calculator, using your real numbers, including ApoB and Lp(a) from the lab guide. Free, and it is the single most decision-relevant thing in this category.
2. **Ask which prevention you are in.** Secondary prevention and true primary prevention are different conversations with different evidence, and you are entitled to be told which one applies to you.
3. **Ask for absolute numbers, not relative ones.** "Reduces risk by 25%" is a relative figure. Ask what it means for someone with your risk over ten years, in whole people. Guidelines increasingly recognize that risk-based selection alone does not capture who actually benefits, and that individualized benefit estimates are better suited to shared decision-making (PMID 40694580).
4. **If muscle symptoms are your barrier**, ask about a structured rechallenge, a different statin, or a lower dose, because the evidence above says that conversation has good odds.
5. **Do everything in the diet guide regardless.** It is the part that helps at every risk level and it is not in competition with anything.

What this guide will not do is tell you to stop a prescription. If you are in secondary prevention, the evidence is strong and stopping is genuinely dangerous. If you are in primary prevention and you and your physician decide the absolute benefit does not justify it for you, that is a legitimate, evidence-consistent decision to make **together**, and it is a completely different act from quitting on the strength of a wellness article.

What outranks everything you can buy

The diet guide covers these in full and they are worth naming here:

- **The eating pattern**, where a network meta-analysis found comprehensive lifestyle modification, aerobic exercise, isometric training, low-sodium and high-potassium salt, salt restriction, breathing-control, meditation, and low-calorie diet all showing clear effects on blood pressure reduction (PMID 32975166).
- **Weight, if you carry extra**, particularly for fatty liver, where sustained weight loss is associated with histological improvement (PMID 41804193).
- **Treating sleep apnea**, which drives blood pressure and is commonly missed.

How to read this guide

Test first. ApoB and Lp(a) change what matters here, and the lab guide covers them.

Give lipid changes eight to twelve weeks before retesting, and blood pressure four weeks.

One change at a time.

Tier 1: red yeast rice, and the thing nobody tells you about it

This is the most requested supplement in this category and it needs the most honest handling.

What the evidence shows on efficacy and safety. A systematic review and meta-analysis of randomized trials found red yeast rice produced a reduction of about **1.02 mmol/L in LDL cholesterol**, which is roughly 39 mg/dL and is a genuinely substantial effect, while concluding that **safety is uncertain** and that it might be a safe and effective option for statin-intolerant patients only once the mild adverse reaction profile is confirmed in studies with adequate safety methodology (PMID 25897793). A later systematic review and meta-analysis focused specifically on safety concluded that red yeast rice used as a lipid-lowering dietary supplement seems overall tolerable and safe in moderately hypercholesterolaemic subjects (PMID 30844537).

Now the part that is usually left out, and it is the most important sentence in this guide.

The active compound in red yeast rice is **monacolin K, which is chemically identical to lovastatin**, a prescription statin.

So if you are taking red yeast rice specifically in order to avoid taking a statin, **you are taking a statin.** You are taking it at a dose that varies between products and batches, without prescription oversight, without the liver and muscle monitoring that comes with the prescription, and in a product category where content is not verified before sale.

That is not an argument against red yeast rice. It is an argument for knowing what you are doing. If red yeast rice is the route you and your clinician choose, the same cautions apply as to any statin: it interacts with the same drugs, it warrants the same monitoring, and it should not be combined with a prescription statin.

Cautions. Do not combine with a prescription statin. Do not use in pregnancy or nursing, or with liver disease. It interacts with the same drugs statins do, including some antibiotics, antifungals, and grapefruit. Tell every prescriber you take it, because it will not be on your medication list otherwise. Some products have historically been found contaminated with citrinin, a nephrotoxin, which is a specific reason to insist on third-party testing here.

Tier 2: omega-3, where the form decides the answer

This is the entry where most guides get it wrong by lumping all fish oil together.

What the evidence shows. A review of the major cardiovascular omega-3 trials concludes that **icosapent ethyl, a purified high-dose EPA form, improves cardiovascular outcomes among patients with elevated triglycerides at high cardiovascular risk, but EPA and DHA together does not.** Current guidelines endorse icosapent ethyl in statin-treated patients at high cardiovascular risk with triglycerides above 135 mg/dL (PMID 36562280).

So the honest translation. The cardiovascular outcome benefit belongs to a specific prescription product, in a specific population, on top of a statin. **It does not transfer to the combined EPA and DHA fish oil capsule on the shelf.** If you have high triglycerides and high cardiovascular risk, the conversation to have is about icosapent ethyl with your physician, not about buying more fish oil.

What over-the-counter fish oil is still reasonable for. Lowering triglycerides, and general dietary omega-3 intake. Not for the outcome claim above.

Cautions. Higher doses increase bleeding tendency, which matters with anticoagulants, antiplatelet drugs, and before surgery.

Tier 3: the supporting cast, honestly graded

Soluble fiber. Psyllium and beta-glucan lower LDL modestly and reliably, and they are cheap, safe, and do other useful things. Start low to avoid bloating and take with plenty of water. This is the least glamorous and most defensible item in the guide.

Plant sterols and stanols. Modest additional LDL lowering. Reasonable as an addition.

Magnesium and potassium are covered in the diet guide, because food is the better route and potassium supplements carry real risk in kidney disease and with certain blood pressure medications. **Do not supplement potassium without your clinician**, particularly on an ACE inhibitor, ARB, or potassium-sparing diuretic.

CoQ10 for statin muscle symptoms, where two meta-analyses disagree

This deserves its own section because you will be told it works, and the evidence is genuinely split.

The negative one. A systematic review and meta-analysis assessing whether CoQ10 improves statin-associated myalgia and helps people stay on statins **did not demonstrate benefit** for muscle pain, and found no improvement in the proportion of patients remaining on statin treatment, with a risk ratio of 0.99 (PMID 32179207).

The positive one. A 2025 systematic review and meta-analysis of CoQ10 for myopathy in statin-treated patients found a reduction in muscle pain, with a weighted mean difference of -0.96 and a confidence interval of -1.88 to -0.03, which is statistically significant but with the upper bound sitting very close to zero. The authors state more research is needed for evidence-based recommendations (PMID 41158831).

How I read that. Two meta-analyses on the same question reaching different conclusions, with the positive one's confidence interval nearly touching no-effect. CoQ10 is safe and inexpensive, and it is a reasonable thing to try if muscle symptoms are your barrier to staying on a statin, which is a worthwhile goal. Just do not expect much, and do not let trying it delay the conversation about rechallenge or a different statin, which has better evidence behind it.

Cautions. CoQ10 can reduce the effect of warfarin. It may modestly lower blood pressure, relevant if you are treated for that.

What I deliberately left out, and why

- **Anything sold as a statin replacement.** Covered above. The blinded rechallenge data say most people who think they cannot tolerate a statin can.
- **Combined EPA and DHA fish oil for cardiovascular outcome prevention.** The outcome benefit belongs to purified high-dose EPA as a prescription, and did not transfer to the combination (PMID 36562280).
- **Niacin at cardiovascular doses.** Outcome trials were disappointing and the side effect burden is real. This is a prescriber conversation, not a shelf purchase.
- **Chelation therapy for atherosclerosis.** Not supported for this use, and not benign.
- **Nattokinase and serrapeptase for "cleaning arteries."** No outcome evidence, and real bleeding-risk questions.
- **High-dose antioxidant vitamins for cardiovascular prevention.** Repeatedly tested, repeatedly disappointing.
- **Anything that encourages you to skip a risk calculation.** Your risk number, produced with a validated calculator and your actual values, is the single most decision-relevant thing in this category, and it is free.

Telling you what to skip is part of what you paid for.

Running a fair trial

Weeks 1 to 4, baseline. Home blood pressure twice daily for a week, waist measurement, and the labs from the lab guide.

Weeks 5 to 16, one change at a time, starting with the diet guide rather than a bottle.

Week 17, retest. Lipids need eight to twelve weeks to reflect a change. Blood pressure responds faster.

Realistic expectations

What is realistic. Modest LDL reduction from soluble fiber and sterols. Meaningful blood pressure change from the diet and movement work. Substantial LDL reduction from red yeast rice, because it contains a statin.

What is not realistic. A supplement replacing appropriate medical treatment in someone at genuine risk. Reversing established plaque with a capsule.

Cost sense. The highest-value items here are a home blood pressure monitor, a tape measure, and psyllium. Together they cost less than one month of most supplement stacks.

When to seek professional care

Emergency: chest pain or pressure, one-sided weakness or speech difficulty, sudden severe breathlessness, tearing chest or back pain. Call 911.

Soon: home blood pressure repeatedly at or above 140/90, chest discomfort on exertion, new breathlessness on stairs, swelling in both ankles, or muscle pain that is severe or comes with dark urine, which needs same-day assessment.

Where to find quality versions of these

Nothing here names a brand, on purpose.

Third-party testing matters more here than almost anywhere. Red yeast rice is the specific case: monacolin content varies between products, and citrinin contamination has been documented. If a red yeast rice product cannot show you independent testing, do not take it.

Match the form to the evidence. Purified high-dose EPA is a prescription, not a shelf product. Psyllium and beta-glucan are the studied soluble fibers.

If you would rather not evaluate all of that yourself, I keep a screened dispensary through Fullscript, which ships directly to you. I hold no physical inventory. It is screened for the standards above.

> ***Root Health dispensary: us.fullscript.com/welcome/drseay***
>
> *Disclosure: Root Health earns a portion of dispensary purchases. That revenue supports the practice, and it is also why this guide was written to be complete and usable on its own. Nothing here was included, excluded, ranked, or dosed based on what it earns, and the three highest-value items in this guide, a blood pressure cuff, a tape measure, and taking your prescribed medication, earn us nothing.*

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your numbers, your calculated risk, and your medication list.

> ***Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=heart-supplement***

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

> ***Get in touch: rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=heart-supplement***

If you would rather start on your own, start with home blood pressure and a waist measurement:

> ***Free tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=heart-supplement***

Read the companion documents alongside this one. **Understanding Your Heart and Metabolic Labs** covers ApoB and Lp(a), and **The Heart and Metabolic Diet Guide** covers the eating pattern that outperforms most of this page.

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